# Unbiased exact diagonalization of quantum spin ice in the $\pi$-flux regime: a zero-transport projector for finite-size gauge contamination

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**Abstract**

The thermodynamics and dynamics of quantum spin ice (QSI) are governed by an emergent compact $U(1)$ gauge field. Its unfrustrated 0-*flux* sector ($J_\pm > 0$) is accessible to sign-problem-free quantum Monte Carlo (QMC); the frustrated $\pi$-*flux* sector ($J_\pm < 0$) realized in the dipolar–octupolar cerium pyrochlores carries a QMC sign problem, leaving exact diagonalization (ED) as the only unbiased finite-temperature method. ED, however, has a finite-size problem specific to QSI: a periodic cluster hosts short ice-preserving ring exchanges that wind the torus and have no infinite-lattice analogue. The dominant ones are four-site loops with coupling $g_4 = 4J_\pm^2/J_{zz}$, which is *second* order in $J_\pm$ and therefore parametrically larger than the genuine photon coupling $g_{\text{hex}} = 12|J_\pm|^3/J_{zz}^2$; they contaminate exactly the emergent-gauge observables one wants (the low-$T$ specific-heat peak and the low-energy $S^{zz}(\mathbf{q}, \omega)$). We remove this contamination with a *zero-transport projector*: retain only ice-manifold processes whose net emergent-charge transport $\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N}\cdot\mathbf{L}$ vanishes modulo the cluster periods. We prove, both mathematically and by direct computation, that the projector (i) annihilates every winding ring exchange at all orders in $J_\pm$, (ii) preserves every contractible ring exchange at full strength, and (iii) coincides with the operator-level twist average taken in the translationally-invariant gauge – clarifying why the naive twist averages used in practice fail (the boundary gauge removes only half; averaging observables rather than the Hamiltonian keeps the full spurious scale, gauge-invariantly). On the 16- and 32-site clusters the projector moves the low-$T$ specific-heat peak and the low-energy $S^{zz}$ weight from the artifact scale $g_4$ to the photon scale $g_{\text{hex}}$, for both flux signs. This makes a 32-site pyrochlore an unbiased laboratory for the $\pi$-flux gauge sector, to be benchmarked against 0-flux QMC and applied at the exchange parameters of the cerium pyrochlores.

## 1 The problem

**Quantum spin ice and its emergent photon.** The minimal model of QSI is the XXZ antiferromagnet on the pyrochlore lattice, written in the local $\langle 111 \rangle$ frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} \big( S_i^+ S_j^- + S_i^- S_j^+ \big), \qquad 0 < |J_\pm| \ll J_{zz}. \tag{1}$$

For $J_\pm \to 0$ the ground manifold is the exponentially degenerate set of 2-in-2-out (ice-rule) states. A small $J_\pm$ lifts this degeneracy through virtual ring exchanges and generates the deconfined $U(1)$ lattice gauge theory of QSI: $S^z$ is the emergent electric field, the third-order hexagonal ring exchange

$$g_{\text{hex}} = \frac{12\,|J_\pm|^3}{J_{zz}^2} \tag{2}$$

is the Maxwell term whose quantum = the emergent photon, and ice-rule defects are gapped, fractionalized spinons (pair threshold $\Delta_\mathrm{s} \approx 0.8\, J_{zz}$ on the clusters below). The *sign* of the ring coupling follows $\mathrm{sgn}(J_\pm^3)$: $J_\pm > 0$ gives the unfrustrated 0-flux vacuum, $J_\pm < 0$ the frustrated $\pi$-flux vacuum. Thermodynamically the photon appears as a low-temperature Schottky peak in $C(T)$ at $T \sim g_\mathrm{hex}$, far below the spinon (charge) peak at $T \sim J_{zz}$; dynamically it appears as low-energy spectral weight in the non-spin-flip channel $S^{zz}(\mathbf{q}, \omega)$.

**Why $\pi$-flux needs exact diagonalization.** In the $S^z$ basis the off-diagonal matrix elements of eq. (1) are $-J_\pm$. For $J_\pm > 0$ they are negative and worldline/stochastic-series QMC is sign-problem-free: the 0-flux QSI is quantitatively characterized. For $J_\pm < 0$ the same elements are positive and QMC acquires a sign problem. The $\pi$-flux regime – believed to describe the cerium pyrochlores $Ce_2Zr_2O_7$, $Ce_2Sn_2O_7$, $Ce_2Hf_2O_7$ – is thus a QMC dead zone, and unbiased finite-temperature information must come from exact diagonalization of a finite cluster.

**The finite-size problem.** A periodic cluster is a flat torus, and its periodic identifications generate ice-preserving ring exchanges that *wind* the torus and do not exist on the infinite lattice. The shortest and most damaging are *four-site* loops. Because a four-site loop factorizes into two disjoint nearest-neighbour (NN) hops, it is generated at *second* order in $J_\pm$,

$$g_4 = \frac{4\, J_\pm^2}{J_{zz}}, \qquad \text{so} \qquad \frac{g_4}{g_\mathrm{hex}} = \frac{J_{zz}}{3|J_\pm|} \gg 1. \tag{3}$$

The artifact is thus *parametrically larger* than the physics, and it grows as $|J_\pm| \to 0$ – exactly the weak-coupling regime of the cerium materials. It lives entirely in the ice manifold, so it contaminates precisely the emergent-gauge observables: it pushes the low-$T$ $C(T)$ peak and the low-energy $S^{zz}$ weight up to the scale $g_4$ instead of $g_\mathrm{hex}$ (quantified in section 6). The spinon continuum ($S^{+-}$ at $\omega \sim J_{zz}$) lies outside the ice manifold and is *not* contaminated.

**Goal.** Remove this finite-size contamination from the gauge sector of a periodic-cluster ED, rigorously and verifiably, so that the 32-site pyrochlore delivers unbiased $\pi$-flux thermodynamics and dynamical structure factor.

## 2 The effective gauge Hamiltonian and the winding loops

**Cluster torus.** The cluster carries $N$ sites with periodic identifications generated by the rows of an integer matrix $\mathbf{L} \in \mathbb{Z}^{3\times3}$ (in units of the cubic lattice constant $a$); position space is the flat torus $T^3 = \mathbb{R}^3/\mathbf{L}\mathbb{Z}^3$ with first homology $\mathbb{Z}^3$. Each NN bond $(i,j)$ has a *wrap vector* $\mathbf{n}_{ij} = \mathrm{round}[(\mathbf{r}_j - \mathbf{r}_i)\mathbf{L}^{-1}]$, zero in the interior and nonzero across the boundary.

**Effective Hamiltonian.** Degenerate ($J_\pm$-)perturbation theory in the ice manifold yields $H_\mathrm{eff} = \sum_\mathcal{C} c_\mathcal{C}\, O_\mathcal{C}$, a sum of ring-exchange operators $O_\mathcal{C} = (S_a^+ S_b^- S_c^+ S_d^- \cdots)$ over ice-preserving cycles $\mathcal{C}$; because each intermediate resolvent is Ising ($\propto J_{zz}$), the coefficient $c_\mathcal{C}$ is exactly $\propto J_\pm^{|\mathcal{C}|/2}$. On the 2×2×2 FCC cluster ($N = 32$) the ice manifold has dimension 2970, so $H_\mathrm{eff}$ is *fully diagonalizable* and its thermodynamics is numerically exact. Enumeration (Schrieffer–Wolff with per-process bookkeeping) gives the census in table 1: the winding four-loops and wrapping hexagons are the contamination; the contractible hexagons carry the photon.

Table 1: Census of ice-preserving ring exchanges on the two production clusters. "winding/wrapping" loops close only through the periodic boundary ($\mathbf{w} \neq \mathbf{0}$); "contractible" hexagons are genuine bulk plaquettes ($\mathbf{w} = \mathbf{0}$). The last column is the survival fraction under the zero-transport projector of section 3.

| cluster | channel | count | order | kept by projector |
|---|---|---|---|---|
| 1×1×1 cubic ($N$=16) | winding four-loops | 36 | $J_\pm^2$ | 0 |
| | wrapping hexagons | 48 | $J_\pm^3$ | 0 |
| | contractible hexagons | 16 | $J_\pm^3$ | 1 |
| 2×2×2 FCC ($N$=32) | winding four-loops | 24 | $J_\pm^2$ | 0 |
| | wrapping hexagons | 96 | $J_\pm^3$ | 0 |
| | contractible hexagons | 32 | $J_\pm^3$ | 1 |

# 3 The zero-transport projector

The gauge-invariant label that distinguishes a genuine (bulk) ring exchange from a torus-winding one is not homology but *charge transport.*

**Definition 1** (Net charge transport)**.** For an ice-manifold process taking configuration $c$ to $c'$ by raising a site set $R$ and lowering a site set $L$ through hops of total wrap $\mathbf{N} = \sum \mathbf{n}$, the net transport is

$$\boldsymbol{\delta} = \boldsymbol{\rho} + \mathbf{N}\cdot\mathbf{L}, \qquad \boldsymbol{\rho} = \sum_{i\in R} \mathbf{r}_i - \sum_{j\in L} \mathbf{r}_j, \tag{4}$$

the physical displacement of the transported $S^z$ (boson) density. $\boldsymbol{\rho}$ is the intrinsic dipole of the flipped-site set (home positions); $\mathbf{N}\cdot\mathbf{L} = \sum_k N_k \mathbf{L}_k$ is the wrap contribution.

**Definition 2** (Zero-transport projector)**.** $\mathcal{P}$ retains exactly the processes with $\boldsymbol{\delta} \equiv \mathbf{0} \pmod{\mathbf{L}}$, i.e. $\boldsymbol{\delta} \in \mathbf{L}\mathbb{Z}^3$, and deletes the rest.

Two implementations, cross-validated in section 5: *(i)* in perturbation theory, a row filter on the enumerated processes; *(ii)* nonperturbatively, a mask on the exactly downfolded ice-manifold Hamiltonian, keeping matrix element $(c, c')$ iff its flip-set dipole $\boldsymbol{\rho} \equiv \mathbf{0} \pmod{\mathbf{L}}$. Because $\mathbf{N}\mathbf{L} \in \mathbf{L}\mathbb{Z}^3$ automatically, the criterion $\boldsymbol{\delta} \in \mathbf{L}\mathbb{Z}^3$ reduces to $\boldsymbol{\rho} \in \mathbf{L}\mathbb{Z}^3$: *the projector's action on an operator depends only on its dipole, at every order.*

# 4 Proof that the projector works: mathematics

## 4.1 Ring exchanges are sums over virtual paths

The naive expectation is that a winding ring exchange picks up its homology holonomy $e^{-i\mathbf{w}\cdot\boldsymbol{\varphi}}$ under a boundary twist $\boldsymbol{\varphi}$ and so is removed by averaging over $\boldsymbol{\varphi}$. This is false, and the reason underlies everything.

**Theorem 1** (Matched-pair factorization)**.** *Let $\mathcal{C}$ be an ice-preserving simple cycle of length $2m$ with winding $\mathbf{w}$. At the leading order at which $O_\mathcal{C}$ appears, its coefficient in the boundary-gauge twisted $H_{\text{eff}}(\boldsymbol{\varphi})$ is*

$$c_\mathcal{C}(\boldsymbol{\varphi}) = -\frac{g_\mathcal{C}}{M} \sum_{\mu=1}^{M} e^{-i\mathbf{N}_\mu\cdot\boldsymbol{\varphi}}, \qquad \sum_\mu \mathbf{N}_\mu = \mathbf{w}\ (M{=}2), \tag{5}$$

*where $\mu$ runs over the $M \geq 2$ perfect matchings of $\mathcal{C}$ into disjoint NN dimer moves and $\mathbf{N}_\mu$ is the total wrap of matching $\mu$. For a two-matching loop the gauge-invariant magnitude is*

$$|c_{\mathcal{C}}(\boldsymbol{\varphi})| = g_{\mathcal{C}} \big|\cos(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\varphi})\big|. \tag{6}$$

*Proof.* At leading order $O_{\mathcal{C}}$ is generated only by flipping each of its $2m$ spins once, i.e. by a product of $m$ disjoint NN dimer moves – a perfect matching of the cycle; an even cycle has $M \geq 2$. Each hop $S_i^+ S_j^-$ carries $e^{-i\mathbf{n}_{ij}\cdot\boldsymbol{\varphi}}$, so matching $\mu$ carries $e^{-i\mathbf{N}_\mu\cdot\boldsymbol{\varphi}}$ with $\mathbf{N}_\mu = \sum_{\text{hops}} \mathbf{n}$. The $M$ matchings share amplitude $g_{\mathcal{C}}/M$ (equal denominators and weight). For a 4-cycle $abcd$ the two matchings $\{ab, cd\}$ and $\{bc, da\}$ together traverse each bond once, so $\mathbf{N}_1 + \mathbf{N}_2 = \mathbf{w}$; (6) follows from $|1 + e^{-i\mathbf{w}\cdot\boldsymbol{\varphi}}| = 2|\cos(\frac{1}{2}\mathbf{w}\cdot\boldsymbol{\varphi})|$. □

The operator thus lives in *several* transport sectors at once; the holonomy is only the *relative* phase of the matchings, and the magnitude (6) *vanishes* at $\mathbf{w}\cdot\boldsymbol{\varphi} = \pi$ – which a single homology phase (constant magnitude) never does.

## 4.2 The projector removes winding, keeps contractible

**Theorem 2** (Transport selection)**.** *Let $\ell_0 = \min_{\mathbf{w}\neq\mathbf{0}} |\mathbf{w}\cdot\mathbf{L}|$ be the shortest nonzero cluster period. If every ring exchange under consideration has alternating dipole $|\boldsymbol{\rho}_{\mathcal{C}}| < \ell_0$, then $\mathcal{P}$ deletes every loop with $\boldsymbol{\rho}_{\mathcal{C}} \neq \mathbf{0}$ at all orders in $J_\pm$ and keeps every loop with $\boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$ at full strength; on the production clusters these are exactly the winding and the contractible loops respectively.*

*Proof.* By the remark below eq. (4), $\mathcal{P}$ keeps $O_{\mathcal{C}}$ iff $\boldsymbol{\rho}_{\mathcal{C}} \in \mathbf{L}\mathbb{Z}^3$, at every order. Since $|\boldsymbol{\rho}_{\mathcal{C}}| < \ell_0$, the only lattice vector it can equal is $\mathbf{0}$: hence $\boldsymbol{\rho}_{\mathcal{C}} \in \mathbf{L}\mathbb{Z}^3 \iff \boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$, and $\mathcal{P}$ keeps $O_{\mathcal{C}}$ iff $\boldsymbol{\rho}_{\mathcal{C}} = \mathbf{0}$.

A genuine bulk plaquette (regular [111]-hexagon, or its four-tetrahedron analogue) is centrosymmetric – its raised and lowered sites share a centroid – so $\boldsymbol{\rho} = \mathbf{0}$ and it is kept. A winding loop has $\boldsymbol{\rho} \neq \mathbf{0}$: were $\boldsymbol{\rho} = \mathbf{0}$, then in the two-matching factorization $\boldsymbol{\delta}_1 + \boldsymbol{\delta}_2 = \mathbf{w}\cdot\mathbf{L}$ with $|\mathbf{w}\cdot\mathbf{L}| \geq \ell_0$ would force some $|\boldsymbol{\delta}_\mu| \geq \frac{1}{2}\ell_0$, yet $\boldsymbol{\delta}_\mu = \boldsymbol{\rho} + \mathbf{N}_\mu\mathbf{L}$ with $\boldsymbol{\rho} = \mathbf{0}$ makes $\boldsymbol{\delta}_\mu$ a lattice vector, hence $\mathbf{0}$ or $\geq \ell_0$ – contradiction. So winding loops are deleted. □

On the production clusters the hypothesis holds with room to spare: the alternating dipole of *every* four-loop and wrapping hexagon has the same magnitude $|\boldsymbol{\rho}| = a/\sqrt{2}$, below $\ell_0 = a$ (cubic) and $\sqrt{2}\,a$ (FCC), while every contractible hexagon has $\boldsymbol{\rho} = \mathbf{0}$ identically – giving the 0/0/1 survival pattern of table 1. The single residual bias is honest: a *contractible* process with a nonzero dipole ($\boldsymbol{\rho} \neq \mathbf{0}$, possible only for non-centrosymmetric loops at order $J_\pm^4$ and beyond) would also be removed; within the NN model these are $O(J_\pm^4/J_{zz}^3)$, parametrically below $g_{\text{hex}}$.

## 4.3 Relation to twist averaging: why the naive versions fail

**Theorem 3** (Twist-average identity)**.** *In the translationally-invariant ("smooth") gauge – a uniform vector potential $\mathbf{A} = \mathbf{L}^{-1}\boldsymbol{\varphi}$ placing $\theta_{ij} = \mathbf{A}\cdot\mathbf{d}_{ij}$ on* every *bond – each process carries $e^{-i\mathbf{A}\cdot\boldsymbol{\delta}}$, and the operator-level twist average over the extended period is $\mathcal{P}$:*

$$\overline{H_{\text{eff}}}^{\text{smooth}} = \int H_{\text{eff}}(\boldsymbol{\varphi})\,\frac{d\boldsymbol{\varphi}}{(\textit{period})} = \mathcal{P}\,H_{\text{eff}}. \tag{7}$$

*Proof.* A hop $i \to j$ contributes $\mathbf{A}\cdot\mathbf{d}_{ij}$; summed along a process the phase is $\mathbf{A}\cdot(\sum\mathbf{d}) = \mathbf{A}\cdot\boldsymbol{\delta}$ with $\boldsymbol{\delta}$ its transport. Averaging $e^{-i\mathbf{A}\cdot\boldsymbol{\delta}} = e^{-i\boldsymbol{\varphi}\cdot(\mathbf{L}^{-1}\boldsymbol{\delta})}$ over the period on which the smooth-gauge $H_{\text{eff}}$ is single-valued gives $\delta_{\mathbf{L}^{-1}\boldsymbol{\delta}\in\mathbb{Z}^3} = \delta_{\boldsymbol{\delta}\in\mathbf{L}\mathbb{Z}^3}$, i.e. $\mathcal{P}$. □

**Remark 1** (The extended period is physical)**.** The averaging period is *not* $[0, 2\pi)^3$. The emergent charges sit at the pyrochlore sublattice positions, which are quarter-integer in the cubic cell, so a transport $\boldsymbol{\delta}$ is quantized in units of $a/4$ and the smooth-gauge phase $e^{-i\boldsymbol{\varphi}\cdot(\mathbf{L}^{-1}\boldsymbol{\delta})}$ is $\boldsymbol{\varphi}$-periodic only modulo $8\pi$ (equivalently, the flux quantum for the fractional emergent charge is larger than the naive boundary twist quantum $2\pi$). Averaging only over $[0, 2\pi)^3$ does *not* project: it leaves a residual $\sim 40\%$ of each winding coupling (section 5). The full period is required, and over it the smooth-gauge operator average equals $\mathcal{P}$ exactly.

So the projector *is* twist averaging, done correctly. This pins down the two ways the naive procedure fails, both verified numerically in section 5:

**Corollary 1** (Failure modes)**.** (a) Wrong gauge. *In the boundary-bond gauge, a winding loop's zero-wrap matching (*$\mathbf{N}_\mu = \mathbf{0}$*) is phase-free, so the operator average keeps* $\frac{1}{2}g_{\mathcal{C}}$ *rather than* 0*; the "half" is a gauge artifact.* (b) Wrong object. *Averaging an* observable $\mathcal{O}(g)$ *(a specific* $C(T)$ *or* $S(\mathbf{q}, \omega)$*) rather than the Hamiltonian keeps every* even *power of the gauge-invariant* $|c_{\mathcal{C}}(\boldsymbol{\varphi})| = g_{\mathcal{C}}|\cos(\frac{1}{2}\mathbf{w}\cdot\boldsymbol{\varphi})|$*, because* $\overline{|c|^{2k}} \neq 0$*; the spurious scale survives in* any *gauge. Since practice averages the measured quantity, naive twist averaging fails.*

## 5 Proof that the projector works: computation

Each mathematical claim is checked exactly on the 16-site cluster (where full ED is available) and, where relevant, on the 32-site FCC cluster.

**(1) Matched-pair magnitude** (6)**.** For a winding four-loop [sites $(0, 2, 8, 10)$, $\mathbf{w} = (-1, 0, -1)$] the exact second-order coefficient obeys, along $\mathbf{w}$,

$$|c(\boldsymbol{\varphi})|/g_4 = 1.000,\ 0.866,\ 0.707,\ 0.500,\ 0.000, \cdots = |\cos(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\varphi})| \text{ to all digits,}$$

vanishing at $\mathbf{w}\cdot\boldsymbol{\varphi} = \pi$. A single homology phase would keep $|c| = g_4$ for all $\boldsymbol{\varphi}$; the vanishing is the two-matching interference of theorem 1.

**(2) Transport selection** (2)**.** Direct enumeration confirms table 1: on both clusters *every* winding four-loop and wrapping hexagon has $\boldsymbol{\rho} \notin \mathbf{L}\mathbb{Z}^3$ (deleted), while *every* contractible hexagon has $\boldsymbol{\rho} = \mathbf{0}$ (kept) – survival 0/0/1, to machine precision, at orders $J_\pm^2$ and $J_\pm^3$.

**(3) Twist-average identity** (3) **and failure modes.** The coefficient of the winding four-loop under the operator-level twist average is

$$-g_4 \text{ (bare)} \;\longrightarrow\; -\tfrac{1}{2}g_4 \text{ (boundary gauge)} \;\longrightarrow\; 0 \text{ (smooth gauge} = \mathcal{P}\text{)},$$

while a contractible hexagon is $+g_{\text{hex}}$ in all three – confirming both theorem 3 and corollary 1(a). For corollary 1(b), the corner-averaged *observable* keeps the spurious scale (section 6).

**(3′) The smooth gauge built explicitly, and the extended period.** Constructing the translationally-invariant gauge on every bond and averaging the *Hamiltonian* over $\boldsymbol{\varphi}$ realizes $\mathcal{P}$ directly, provided the period of remark 1 is used. At $J_\pm = -0.1$ the winding four-loop coefficient and the resulting low-$T$ $C(T)$ peak are

Table 2: Low-$T$ specific-heat (gauge) peak $T_{\rm peak}/g_{\rm hex}$ on the 32-site FCC cluster (exact full spectrum of the 2970-state ice manifold, order $J_\pm^4$). Bare periodic ED sits at the artifact scale; $\mathcal{P}$ restores the photon scale $O(1)$.

| $J_\pm/J_{zz}$ | bare PBC | $\mathcal{P}$ | $g_4/g_{\rm hex}$ |
|---|---|---|---|
| $-0.03$ | 5.4 | 0.6 | 11.1 |
| $-0.05$ | 3.1 | 0.3 | 6.7 |
| $+0.046$ (0-flux) | 10.1 | 0.8 | 7.2 |

| smooth-gauge average | winding coeff. $/g_4$ | $T_{\rm peak}/g_{\rm hex}$ |
|---|---|---|
| none (bare) | $-1.00$ | 3.18 |
| over $[0,2\pi)^3$ | $+0.40$ | 0.84 |
| over $[0,8\pi)^3$ (extended) | 0.00 | 0.62 |
| $\mathcal{P}$ (reference) | 0.00 | 0.62 |

The extended-period smooth-gauge average agrees with $\mathcal{P}$ to all digits (peak 0.00748 $J_{zz}$ either way); the naive $[0,2\pi)^3$ average leaves the peak dirty. This is the explicit, gauge-covariant demonstration that "twist averaging throughout the bulk" works and equals the projector once the emergent-charge period is respected.

**(4) Nonperturbative agreement.** The dipole mask on the exactly downfolded 16-site Hamiltonian reproduces the perturbative $\mathcal{P}$: e.g. the low-$T$ $C(T)$ peak at $J_\pm = -0.05$ agrees to 0.00122 vs. 0.00127 $J_{zz}$, validating the truncation.

## 6 Demonstration: the artifact removed, both flux sectors

**Thermodynamics.** Table 2 lists the low-$T$ (gauge) peak of $C(T)$ in units of $g_{\rm hex}$. Bare periodic ED places it at the artifact scale, drifting up as $g_4/g_{\rm hex} = J_{zz}/3|J_\pm|$ toward weak coupling; the projector brings it to $O(g_{\rm hex})$ for both flux signs. Corner-averaging the *observable* barely moves it (corollary 1(b)).

**Dynamical structure factor.** The emergent photon is the low-energy weight of the non-spin-flip channel $S^{zz}(\mathbf{q},\omega) = \sum_n |\langle n|S^z_{\mathbf{q}}|0\rangle|^2\delta(\omega-\omega_n)$; since $S^z$ is diagonal in the ice basis this is computed exactly from the projected $H_{\rm eff}$. Table 3 gives the low-$\omega$ weight centroid: bare ED places it at the artifact scale $g_4$, the observable twist-average leaves it there, and only $\mathcal{P}$ brings it to the photon scale. (The spinon continuum $S^{+-}$ at $\omega \sim J_{zz}$ is uncontaminated and is obtained separately by full-Hilbert-space methods.)

Both tables make the same point in the two experimentally relevant observables: the finite-size artifact sits at $g_4$, twist averaging does not remove it, and $\mathcal{P}$ restores the photon scale $g_{\rm hex}$ for both 0- and $\pi$-flux.

## 7 Outlook: the $\pi$-flux campaign

The projector makes the 32-site pyrochlore an unbiased calculator of the QSI gauge sector, and the couplings scale as $J_\pm^k$ exactly, so a single effective-Hamiltonian build serves every coupling

Table 3: Low-$\omega$ centroid of the emergent-photon structure factor $S^{zz}$ (in units of $g_4$) at $J_\pm = -0.05\,J_{zz}$. Twist averaging of the observable fails; the projector reaches the photon scale $\sim g_{\rm hex} = 0.15\,g_4$.

| scheme | 16-site | 32-site FCC |
|---|---|---|
| bare PBC | $4.2\,g_4$ | $1.8\,g_4$ |
| twist-averaged observable (8 corners) | $2.6\,g_4$ | – |
| zero-transport projector $\mathcal{P}$ | $0.47\,g_4$ | $0.28\,g_4$ |

and both signs. The program is: *(A)* benchmark the projected 0-flux $C(T)$, $S(T)$ and photon $S^{zz}$ against sign-problem-free QMC at a common coupling; *(B)* cross into $\pi$-flux and compute the same quantities where QMC cannot, resolving the emergent-photon lower peak; *(C)* evaluate at the measured exchange parameters of $Ce_2Zr_2O_7$, $Ce_2Sn_2O_7$ and $Ce_2Hf_2O_7$ and confront the neutron $S(\mathbf{q},\omega)$ and the specific heat directly. The uncontaminated spinon continuum is supplied by full-Hilbert-space finite-temperature Lanczos, with twist averaging used for its legitimate purpose there – Brillouin-zone interpolation of the genuine bands, not contamination removal.